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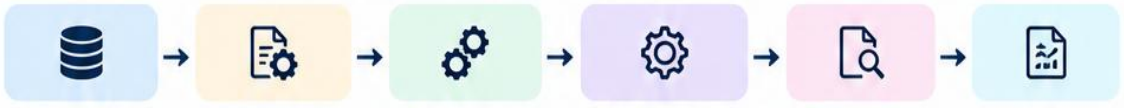
PRACTICAL ASSIGNMENT REPORT

AWS CLOUD & DATA ENGINEERING

Submitted by

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Data Engineering Pipeline



Source Data (Structured Data) → **Data Ingestion** (Read using SQL & Linux) → **Data Storage** (Amazon S3) → **Data Processing & Transformation** (AWS Glue) → **Data Catalog** (AWS Glue Data Catalog) → **Output** (Parquet Format)

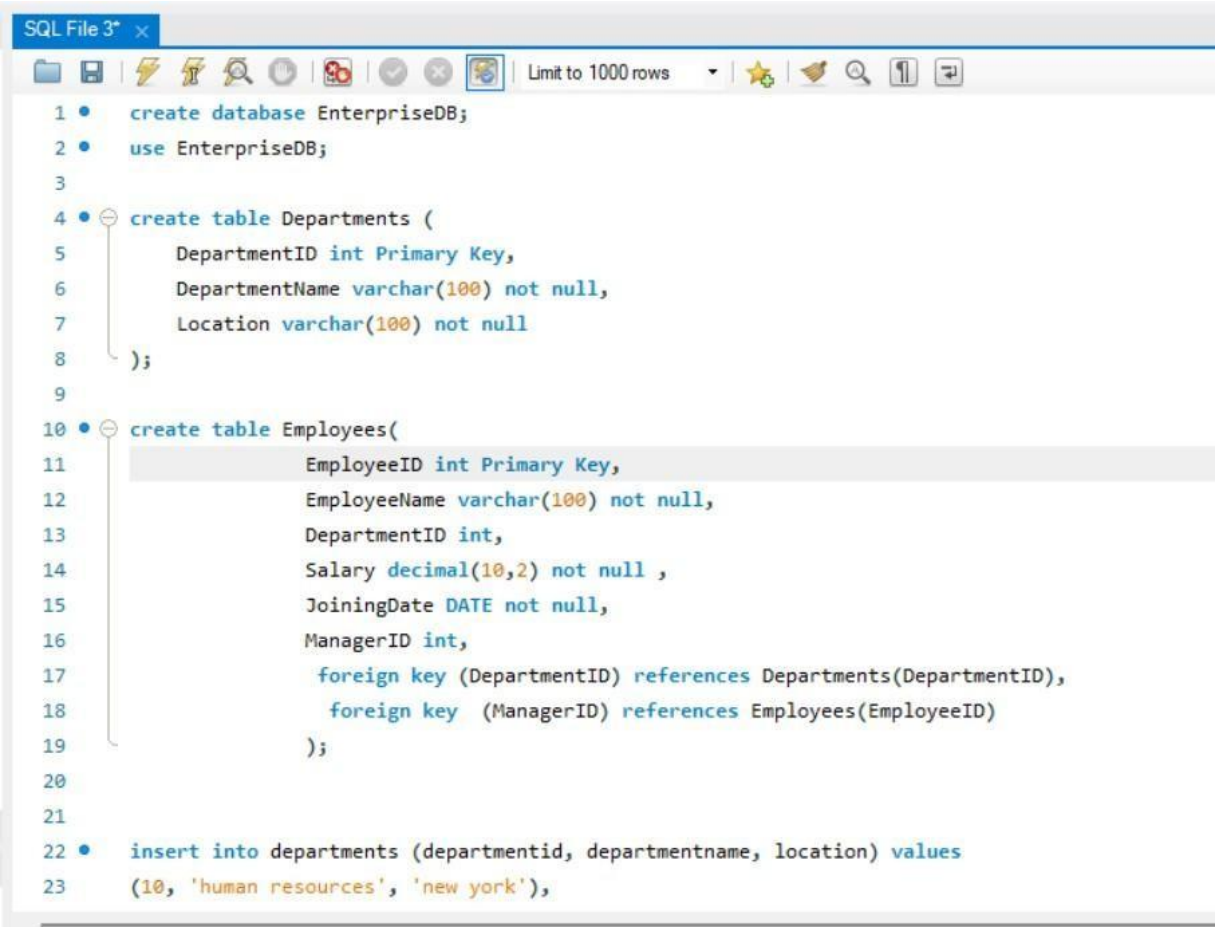
Technologies Used

SQL | Linux | Amazon S3 | AWS IAM | AWS Glue | AWS Glue Data Catalog | AWS Glue Visual ETL | Apache Parquet

Objective

To practice SQL queries and Linux commands, and to build an AWS data engineering pipeline that reads employee data from Amazon S3, discovers its schema using AWS Glue Crawler, transforms the data using AWS Glue Visual ETL, and stores the curated output in Amazon S3 in Parquet format.





a. Display employee names, department names, and manager names using joins. **(2 marks)**

```
SQL File 3*
Limit to 1000 rows
43    (14, 'nathan king', 40, 75000.00, '2023-11-30', 12),
44    (15, 'olivia wright', 40, 60000.00, '2024-03-18', 13);
45
46    -- TASKS
47     -- a. Display employee names, department names, and manager names using joins. (2 marks)
48     select
49        e.employeename as employee,
50        d.departmentname as department,
51        m.employeename as manager
52    from employees e
53    left join departments d on e.departmentid = d.departmentid
54    left join employees m on e.managerid = m.employeeid;
55
56    -- b. Find employees earning more than their department's average salary. (2 marks)
57
58    select e.employeeid, e.employeename, e.departmentid, e.salary
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee	department	manager
alice smith	engineering	NULL
bob jones	engineering	alice smith
charlie brown	engineering	alice smith
david miller	engineering	bob jones
eva green	engineering	bob jones

Result 4

b. Find employees earning more than their department's average salary. **(2 marks)**

```
56    -- b. Find employees earning more than their department's average salary. (2 marks)
57
58    select e.employeeid, e.employeename, e.departmentid, e.salary
59    from employees e
60    where e.salary > (
61        select avg(sub.salary)
62        from employees sub
63        where sub.departmentid = e.departmentid
64    );
65
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employeeid	employeename	departmentid	salary
1	alice smith	20	150000.00
2	bob jones	20	120000.00
3	charlie brown	20	110000.00
7	grace harris	30	140000.00
10	jack walker	10	95000.00
12	leo allen	40	115000.00
13	mia young	40	85000.00

c. Find the second-highest distinct salary without using LIMIT, TOP, or FETCH FIRST. **(2 marks)**

```
65
66      -- c. Find the second-highest distinct salary without using LIMIT, TOP, or FETCH FIRST. (2 marks)
67
68      select max(salary) as secondhighestsalary
69      from employees
70      where salary < (select max(salary) from employees);
71
72      -- d. Rank employees by salary within each department and display the top three employees per depar
73
74      with rankedemployees as (
75          select
76              employeeid,
77              employeename,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

secondhighestsalary
140000.00

d. Rank employees by salary within each department and display the top three employees per department, retaining ties at the cutoff. **(2 marks)**

```sql
where salary < (select max(salary) from employees);

-- d. Rank employees by salary within each department and display the top three employees per de

with rankedemployees as (
    select
        employeeid,
        employeename,
        departmentid,
        salary,
        dense_rank() over (partition by departmentid order by salary desc) as salaryrank
    from employees
)
select employeeid, employeename, departmentid, salary, salaryrank
from rankedemployees
where salaryrank <= 3;


-- Question 2: Sales Analytics and Query Optimization (6 Marks)

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employeeid	employeename	departmentid	salary	salaryrank
10	jack walker	10	95000.00	1
11	karen hall	10	70000.00	2
1	alice smith	20	150000.00	1
2	bob jones	20	120000.00	2
3	charlie brown	20	110000.00	3
7	grace harris	30	140000.00	1
8	henry clark	30	105000.00	2

Result 7

Question 2: Sales Analytics and Query Optimization (6 Marks)

Create a Sales table with the following columns:

- SaleID

- CustomerID
- SaleDate
- ProductName
- Quantity
- UnitPrice

Insert at least 15 sample records covering multiple customers, products, and dates.

Write SQL queries to:

```sql
-- Question 2: Sales Analytics and Query Optimization (6 Marks)

create table sales (
    saleid int primary key,
    customerid int,
    saledate date,
    productname varchar(100),
    quantity int,
    unitprice decimal(10,2)
);
```

a. Calculate total revenue for each product. **(2 marks)**

```sql
-- a. Calculate total revenue for each product. (2 marks)
select
    productname,
    sum(quantity * unitprice) as total_revenue
from sales
group by productname;
```

b. Calculate monthly revenue and month-over-month revenue change using a window function. **(2 marks)**

```sql
-- b. Calculate monthly revenue and month-over-month revenue change using a window function. (2 marks)
select
    sale_year,
    sale_month,
    monthly_revenue,
    monthly_revenue - lag(monthly_revenue) over (
        order by sale_year, sale_month
    ) as revenue_change
from monthly_sales
order by sale_year, sale_month;
```

c. Rewrite the following query to improve the potential for index usage and explain why your version may perform better. **(2 marks)**

SELECT *

FROM Employees

WHERE YEAR(JoiningDate) = 2025; Assume

JoiningDate may be indexed.

Ans :

SELECT *
FROM Employees
WHERE JoiningDate >= '2025-01-01' AND JoiningDate <= '2025-12-31';

Question 3: SQL Troubleshooting (6 Marks)

A company maintains millions of employee and transaction records. Its reporting queries are running slowly. Perform the following tasks:

a. Explain how an execution plan helps identify expensive operations in a query. **(2 marks) :**

Exposes Data Access Methods: It reveals if the database is using a costly **Full Table Scan** (scanning millions of rows) instead of an efficient **Index Seek/Scan**.

Highlights Costly Operations: It attaches a weight or percentage cost to each individual step (e.g., joins, sorts, aggregations), allowing developers to pinpoint the exact operation consuming the most CPU or I/O.

Flags Mismatches & Bottlenecks: It shows actual versus estimated row counts. Large discrepancies indicate stale statistics, which lead to suboptimal join choices (like a nested loop instead of a hash match).

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b. Recommend an appropriate index strategy for queries filtering employees by DepartmentID and Salary. Explain your reasoning. **(2 marks)** :

CREATE INDEX idx_emp_dept_salary ON Employees (DepartmentID, Salary);

c. Write a query to identify duplicate customer email addresses and display only addresses occurring more than once. **(2 marks)**

```
SELECT EmailAddress, COUNT(EmailAddress) AS OccurrenceCount
FROM Customers
GROUP BY EmailAddress
HAVING COUNT(EmailAddress) > 1;
```

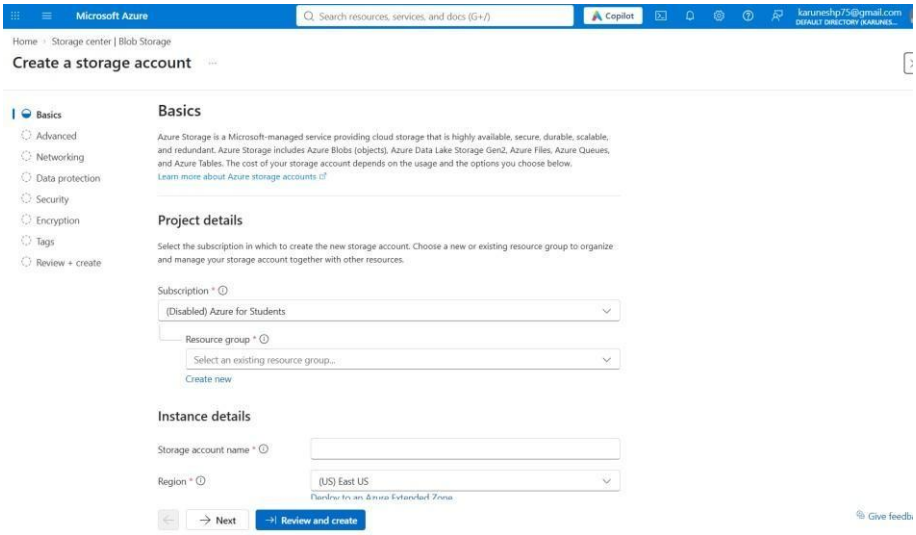
SECTION F: MICROSOFT AZURE || Maximum Marks: 15

Question 15: Azure Data Lake Storage and Security (5 Marks)

A company needs to store confidential business data in Azure for analytics.

Perform or document the following:

a. Configure or describe a storage account with hierarchical namespace enabled for Azure Data Lake Storage Gen2. **(1 mark)**



b. Create a filesystem or container and upload a sample dataset. **(1 mark)**

c. Explain how Microsoft Entra ID and Azure role-based access control can manage access to the data. **(2 marks)** :

Identity Authentication via Entra ID: All users, security groups, or application service principals must authenticate their identity securely using Microsoft Entra ID tokens instead of shared access keys.

Control-Plane Management (Azure RBAC): Azure RBAC roles are applied at the Storage Account or Container level to govern administrative operations. Standard built-in roles include:

Storage Blob Data Owner: Full control over storage container configurations and data.

Storage Blob Data Contributor: Permits reading, writing, and deleting blob/file assets.

Storage Blob Data Reader: Restricts access to read-only capabilities.

d. Explain how you would verify that unauthorized users cannot access the dataset. **(1 mark)** :

1. **Explicit Identity Testing:** Have an unprivileged Entra ID test user or external client attempt to log into the storage account via Azure Storage Explorer, Azure Portal, or a PySpark connection string. The system must immediately throw a 403 Authorization Failed or StorageException: Access Denied error block.
2. **Public Access Verification:** Generate the direct URL endpoint of the uploaded sample file (e.g., https://<account>.dfs.core.windows.net/container/transactions.csv) and attempt to access it through an anonymous browser incognito window. It should return a standard XML/JSON authentication block failure notice, proving anonymous public entry is disabled.
3. **Monitor via Azure Monitor logs:** Inspect **Storage Insights Diagnostic Logs** to verify that failed unauthenticated queries are successfully flagged, categorized, and blocked by the Azure storage security firewall.

Question 16: Azure VM Troubleshooting (5 Marks)

A virtual machine is running in Azure, but administrators cannot connect to it.

Perform or document the following:

a. Verify the VM's provisioning and power states. **(1 mark)**

Provisioning State: This represents the status of the Azure fabric deployment. It should say **Succeeded**. If it says *Failed* or *Updating*, the underlying hardware allocation or extension deployment failed, requiring a redeployment or reapply operation.

Power State: This represents the actual operational runtime state of the hardware. It must display **Running**. If it shows *Stopped* or *Deallocated*, the VM is turned off and cannot accept network traffic.

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b. Explain how Network Security Group rules can prevent connectivity. **(1 mark)**

A **Network Security Group (NSG)** acts as a stateful firewall controlling inbound and outbound traffic. It can block connection attempts through the following mechanics:

Missing Inbound Allow Rules: By default, all inbound traffic from the public internet is blocked by the default **DenyAllInbound** rule. If an explicit rule allowing your connection protocol—such as **Port 22** for Linux SSH or **Port 3389** for Windows RDP—is missing, the connection will drop.

Rule Priority Overrides: NSG rules are evaluated sequentially based on priority numbers (100 to 65000; lower numbers execute first). Even if you have a rule allowing port access at priority 500, a higher-priority rule (e.g., priority 200) that explicitly denies that same IP subnet or port will take precedence and block the traffic.

c. Explain how subnet and routing configuration affect connectivity. **(1 mark)**

Traffic must find a valid path between your client machine and the VM network interface. Connectivity fails if errors exist within the Virtual Network (VNet) routing fabric:

Route Tables & UDRs: Azure automatically provisions system routes to direct traffic. However, if a **User-Defined Route (UDR)** is attached to the VM's subnet that overrides default paths for instance, routing all internet-bound

traffic through a misconfigured or down network virtual appliance (NVA) or firewall the return traffic package gets lost in a routing black hole.

Missing Gateway/Peering Links: If you are trying to access the VM via a private network layout (such as a VPN or ExpressRoute connection), connectivity will drop if **VNet Peering** configurations are broken, or if the subnet's route table lacks a route directing private ranges to the Virtual Network Gateway.

d. Recommend a secure troubleshooting method that avoids unnecessarily exposing the VM to the public internet. **(2 marks)**

Use **Azure Bastion** to establish a secure RDP/SSH session directly inside the Azure Portal browser via TLS/443 without assigning a public IP address to the VM. Alternatively, connect through the native **Serial Console** to troubleshoot lower-level guest OS boot errors over the secure Azure management plane.

Question 17: Azure Data Integration (5 Marks)

A company receives a CSV file containing daily customer transactions and needs to prepare it for analysis.

Design a solution using Azure services.

a. Select an appropriate service to orchestrate data movement. **(1 mark)**

Selected Orchestration Service: Use **Azure Data Factory (ADF)** as the cloud architect to orchestrate data movements. It offers enterprise-grade scheduling, automated workflows, and native serverless connectivity across diverse storage environments.

b. Explain how to copy the CSV file into Azure Data Lake Storage Gen2. **(1 mark)**

Copying to ADLS Gen2: Implement an ADF **Copy Activity** that maps data from your incoming source directory to the destination container. It manages compression, file parsing, and secure, high-throughput cloud loading automatically.

c. Describe how to organize files by ingestion date. **(1 mark)**

I**ngestion Date Organization:** Apply dynamic time expressions within the ADF Sink folder path configurations to generate dated folders on the fly. This automatically segments files into structured paths like year=2026/month=10/day=28/ as they land

d. Explain how to handle malformed records and missing values. **(1 mark)**

Handling Faulty Records: Route data through an ADF **Mapping Data Flow** to clean and process fields before saving. Use *Conditional Split* to isolate malformed rows into a separate error directory and *Derived Column* to substitute missing values with defaults.

e. Explain how to monitor pipeline execution and validate data completeness. **(1 mark)**

Monitoring & Validation: Track runtime pipeline success, pipeline delays, and errors using the visual **ADF Monitor Dashboard** and Azure Monitor logs. Use *Get Metadata* activities to cross-verify row counts and file sizes between source and destination layers.
